

Abdullah Khaled

Robotics & Machine Learning Engineering Student

📍 Frisco, TX ✉️ abdullahkhaledo224@gmail.com ☎️ (469) 347-2884 🔗 akhaled247.github.io 🌐 [akhaled247](https://akhaled247.github.io)
📱 [akhaled247](https://akhaled247.github.io)

Education

Wakeland High School, High School Diploma – Frisco, TX Aug 2023 – May 2027

- Unweighted GPA: 4.0
- 18 AP courses by graduation (completed and planned)

Frisco ISD Career and Technical Education Center, Project Lead The Way – Frisco, TX Aug 2023 – May 2027

- Frisco ISD Career and Technical Education Center Project Lead the Way (PLTW) Nationally Distinguished Engineering Program
- Courses: PLTW Introduction to Engineering Design (9th grade), PLTW Principles of Engineering (10th grade), PLTW Digital Electronics (11th grade), PLTW Engineering Design and Development (12th grade)
- Engineering Portfolio: akhaled247.github.io/pltw-engineering

Technical and Computer Skills

Engineering: CAD/CAM: Fusion 360 (Autodesk), Onshape; NI Multisim, KiCAD, EasyEDA, Soldering

Software: Python, Java, JavaScript, HTML, Jekyll, CSS, Google Workspace, R, Google Sheets, MS Office

Technologies: OpenCV, SolvePnP, TensorFlow, Federated Learning, Kalman Filtering, Finite State Machines, Mechanism Simulation, Control Theory

Engineering Projects

OralVision (Oral Disease Detector) – Conrad Challenge; Diamond Challenge Oct 2025 – present

- Designed oral disease detection system using HIPAA-compliant federated learning with TensorFlow Federated, enabling distributed model training without sharing patient data
- Integrated xiao-esp32-c6 MCU with OV5642 5MP camera sensor, powered by two AAA batteries via a 5V voltage regulator
- Iterated through 3 PCB revisions and 5 3D-printed snap-fit case revisions to improve ergonomics, manufacturability, and reliability
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs
- Conrad Challenge Innovator Award and Alternate Finalist; Diamond Challenge Semi-Finalist
- Patent application in progress with the United States Patent and Trademark Office (USPTO)

WatchFall (Senior Fall Monitor Device) – Samsung Solve for Tomorrow Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Implemented autonomous Bezier path-style motion planning for fall scenarios using a command-based architecture
- Engineered hybrid dataset combining public and custom annotated data, reducing false positive rate

Technical Experience

Chief Technical Advisor, FIRST Robotics Competition (FRC) ITKAN Robotics – Frisco, TX Aug 2025 – present

- Developed components of competition robot software stack integrating localization, path planning, telemetry, and mechanism control in Java
- Built publish-subscribe telemetry and motor logging framework for real-time robot diagnostics and performance analysis
- Led development of robot state estimation and localization system using sensor fusion of AprilTag vision and swerve-drive odometry, achieving <2 inch positional error
- Implemented finite-state machine control architecture with Dijkstra-based path planning using time-weighted graph
- Used WPILib Java mechanism simulation to optimize telescoping arm geometry and motion constraints
- Maintained and extended a >1000-line Java codebase spanning autonomous routines, teleoperated control, vision, and mechanisms using WPILib's command-based framework

Software Lead, FIRST Robotics Competition (FRC) ITKAN Robotics – Frisco, TX Dec 2024 – July 2025

- Provided technical leadership on the organization of software team projects, educating and training new programming team members, and maintaining the team's FRC codebases
- Trained new programming/software team members in Java, robotics-specific coding paradigms, debugging and troubleshooting, and mechanism control using control theory
- Implemented autonomous Bezier pathing using a command-based architecture to coordinate motion profiles, mechanism actions, and sensor feedback
- Advanced team to Texas State Championship playoffs (89 teams)
- Robot codebase and documentation: akhaled247.github.io/first/reefscape

Programming Mentor, ITKAN Robotics of ICF (FTC 26357, 28931) – Frisco, TX Sept 2024 – Feb 2026

- Implemented Kalman-filtered state estimation combining AprilTag vision and wheel odometry for real-time robot localization and automated turret targeting
- Mentored all-rookie FTC team through full engineering design cycle from prototyping to competition
- Mentor refugee students in FLL workshops and lead weekly summer training for FTC and FRC students using self-developed OARobotics curriculum (akhaled247.github.io/personal/oarobotics)

Founder and President, Wolverine Robotics, Wakeland High School – Frisco, TX Aug 2024 – present

- Founded Wakeland Robotics Club, growing membership from 0 to 50+ students in the first year
- Secured funding and constructed competition-ready robot in under 2 weeks, advancing to UIL State Championship
- Teach CAD software (Onshape) and Java programming through monthly technical workshops
- Manage club operations including curriculum design, calendar management, and budget management

Honors & Awards

- (2026) OralVision: Conrad Challenge Innovator Award & Alternate Finalist, Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner, Control Award Winner (FiT-North Dallas Semi-Regional Championship, 25 teams), Control Award Winner (FiT-North U-League Tournament)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist, Event Winner (FIRST® in Texas Victoria, 25 teams); Autonomous Award (FiT Plano, 31 teams); Creativity Award (FiT Amarillo, 36 teams)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) Collegeboard: AP Scholar with Distinction, National Recognition Program: School Recognition Award